Heckscher-Ohlin Model II

Wen-Bin Chuang

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Stolper-Samuelson Theorem

The `Stolper-Samuelson Theorem` (1941) is one of the most important results in the Heckscher-Ohlin model. It describes how `changes in goods prices` affect `real factor incomes`.

Core Idea: An increase in the `relative price` of a good will raise the `real return` to the factor used intensively in that good and lower the real return to the other factor. In simpler terms:

- If the price of the **labor-intensive** good rises → **Real wages rise**, real return to capital falls.
- If the price of the **capital-intensive** good rises → **Real return to capital rises**, real wages fall.

"In the Heckscher-Ohlin world, trade `benefits the abundant factor` and hurts the scarce factor. This is the Stolper-Samuelson Theorem — one of the strongest and most politically relevant results in international trade theory." In autarky, there is a `lower` relative price of the abundant factor; while the trade can raise the relative price. + S-S theorem, their real wages rise.

Core Assumptions (Same as H-O Model)

Assumption	Description
Two countries, Two goods, Two factors	Standard 2×2×2 model
`Perfect Competition`	In both goods and factor markets
Constant Returns to Scale (CRS)	Production functions are homogeneous of degree 1
Identical Homothetic Preferences	Same tastes across countries
Technology	Same technology in both countries
Factors mobile within country	Labor and Capital mobile between sectors
Factors immobile between countries	No international factor mobility
No transport costs / trade barriers	Free trade
Full employment	All factors are fully utilized

Main Mechanism

When a country **opens to trade**:

- **Labor-abundant country** exports the **labor-intensive** good → price of labor-intensive good **increases** (relative to autarky).
- According to Stolper-Samuelson:
  - Real wage of labor (**abundant factor**) **rises**
  - Real return to capital (**scarce factor**) **falls**

This happens because:

- Increased `demand` for the labor-intensive good → increased `demand` for labor → wages rise.
- As resources move into the labor-intensive sector, capital becomes relatively `less demanded` → return to capital falls.

1. **Trade opens** → World prices differ from autarky prices → Domestic relative goods price changes.
2. Firms in the sector with the rising price `expand production`; firms in the other sector `contract`.
3. Expanding sector demands more of its intensive factor; contracting sector releases factors.
4. **Factor markets clear**: Total endowments are fixed, so factor prices must adjust to reallocate factors across sectors.
5. The factor used intensively in the expanding sector faces excess demand → its price rises.
6. The other factor faces excess supply → its price falls.
7. **Magnification**: Factor price changes exceed output price changes, ensuring real returns move unambiguously.

Important Implications

Situation	Effect on Real Wage (w)	Effect on Real Return to Capital (r)
Price of labor-intensive good ↑	**Increases**	Decreases
Price of capital-intensive good ↑	Decreases	**Increases**
Labor-abundant country opens to trade	**Winners**: Workers	**Losers**: Capital owners
Capital-abundant country opens to trade	**Losers**: Workers	**Winners**: Capital owners

Political Economy Implications

Dimension	Prediction
Income Distribution	Trade `widens` the gap between abundant and scarce factors
Political Coalitions	Abundant-factor owners support free trade; scarce-factor owners demand protection
Historical Evidence	19th-century US: Capital-abundant → manufacturers favored tariffs; Land-abundant South favored free trade. Post-WWII Europe: Labor-abundant → unions supported trade liberalization.
Policy Design	Justifies trade adjustment assistance, compensation schemes, progressive taxation

Empirical Evidence & Testing Challenges

- **Developing countries**: `Trade liberalization` often raises demand for `unskilled labor (abundant factor)` → reduces wage inequality in some cases (e.g., Mexico post-NAFTA, though results are mixed).
- **Advanced economies**: Opening to low-wage country imports correlates with relative wage declines for low-skilled workers in manufacturing (consistent with SS if low-skilled labor is scarce relative to developing nations).

Testing Difficulties:

1. **Simultaneity**: Goods prices, technology, and endowments change simultaneously.
2. **Factor heterogeneity**: "Labor" `isn't homogeneous`; skill, education, and region matter.
3. **Incomplete specialization**: Many countries don't fully specialize, blurring price transmission.
4. **Trade vs. Technology**: Skill-biased technological change often confounds SS predictions in advanced economies.

Modern Approaches:

- Industry-level wage regressions controlling for technology and capital accumulation.
- Natural experiments (e.g., tariff reductions, trade agreement implementations).
- Structural estimation of H-O-SS systems (e.g., Hakura & Rodrik, 1999; Goldberg & Pavcnik, 2007).

Key Takeaways

1. **Trade affects factor incomes, not just national income**.
2. **Magnification effect** ensures real returns move unambiguously for abundant/scarcity factors.
3. **Distributional consequences** are central to trade policy design and political feasibility.
4. The theorem is a **comparative static result** about goods price → factor price transmission, applicable beyond trade (e.g., tariffs, subsidies, technological shocks).
5. While empirically nuanced, SS remains the foundational framework for understanding **who wins and who loses from globalization**.

Mathematical Derivation

1. Starting Point: Zero-Profit Conditions (Competitive Markets)

In the `2×2 Heckscher-Ohlin model`:

$$w \cdot a_{LX} + r \cdot a_{KX} = P_X \quad \text{(Good X — labor-intensive)}$$
$$w \cdot a_{LY} + r \cdot a_{KY} = P_Y \quad \text{(Good Y — capital-intensive)}$$

Where:

- w = wage (return to labor), r = rental rate (return to capital)
- a_{ij} = unit input requirements (assumed fixed in the short run for differentiation, or cost-minimizing)

2. Differentiate with Respect to Prices (Percentage Change Form)

Take `total differentials` and divide by the original equations to get **percentage changes** (denoted by hats: $\hat{x} = dx/x$):

$$\theta_{LX}\hat{w} + \theta_{KX}\hat{r} = \hat{P}_X$$
$$\theta_{LY}\hat{w} + \theta_{KY}\hat{r} = \hat{P}_Y$$

Where **cost shares** (θ_{ij}) are defined as:

$$\theta_{LX} = \frac{wa_{LX}}{P_X}, \quad \theta_{KX} = \frac{ra_{KX}}{P_X}, \quad \theta_{LX} + \theta_{KX} = 1$$

(and similarly for good Y). Note that because `X is labor-intensive`: $\theta_{KX} < \theta_{KY}$ and $\theta_{LX} > \theta_{LY}$

3. Matrix Form

$$\begin{bmatrix} \theta_{LX} & \theta_{KX} \\ \theta_{LY} & \theta_{KY} \end{bmatrix} \begin{bmatrix} \hat{w} \\ \hat{r} \end{bmatrix} = \begin{bmatrix} \hat{P}_X \\ \hat{P}_Y \end{bmatrix}$$

Let $\mathbf{\Theta}$ be the cost-share matrix. Then:

$$\begin{bmatrix} \hat{w} \\ \hat{r} \end{bmatrix} = \mathbf{\Theta}^{-1} \begin{bmatrix} \hat{P}_X \\ \hat{P}_Y \end{bmatrix}$$

4. Solve Using Cramer's Rule (Key Result)

The determinant of $\mathbf{\Theta}$ is:

$$|\mathbf{\Theta}| = \theta_{LX}\theta_{KY} - \theta_{LY}\theta_{KX} = \theta_{LX}(1 - \theta_{KY}) - (1 - \theta_{LX})\theta_{KX}$$

Because X is labor-intensive ($\theta_{KX} < \theta_{KY}$), we have $|\mathbf{\Theta}| > 0$. Solving for changes:

$$\hat{w} = \frac{\theta_{KY}\hat{P}_X - \theta_{KX}\hat{P}_Y}{|\mathbf{\Theta}|} \quad \hat{r} = \frac{\theta_{LX}\hat{P}_Y - \theta_{LY}\hat{P}_X}{|\mathbf{\Theta}|} \quad \text{(signs flip due to positive determinant)}$$

5. Main Stolper-Samuelson Results (Percentage Changes)

`Assume that` the relative price of the labor-intensive good rises: $\hat{P}_X > \hat{P}_Y$ (i.e., $\hat{P}_X - \hat{P}_Y > 0$). Then:

- **Real return to labor rises more than proportionally**: $\hat{w} > \hat{P}_X > \hat{P}_Y > \hat{r}$
- **Real return to capital falls**: $\hat{r} < 0$ in real terms (relative to both goods)

Magnification Effect: The change in factor prices is *magnified* relative to goods price changes:

$$|\hat{w}| > |\hat{P}_X| > |\hat{P}_Y| > |\hat{r}|$$

(with opposite directions for the two factors).

6. Intuitive Summary

- An increase in P_X (labor-intensive good) increases `demand for labor more` than for capital → w rises strongly.
- Capital demand relatively falls → r falls (or rises less).
- Because of the magnification effect, real returns to capital fall in terms of *both* goods, and real wages rise in terms of both goods.

Rybczynski Theorem (1955)

The `Rybczynski Theorem` (1955) examines the effect of an increase in `factor endowment` on `output levels` at constant goods prices. It explains how `economic growth` through factor accumulation leads to changes in the production structure — favoring the sector that uses the growing factor intensively. At **constant relative goods prices**, an increase in the endowment of one factor will:

- Increase the output of the good that **intensively uses** that factor (**more than proportionally**),
- Decrease the output of the other good (**absolute decline** possible).

Core Idea: In simple terms: “If a country gets more of one factor (e.g., more labor), it will produce much more of the labor-intensive good and less of the capital-intensive good.” This is the **supply-side dual** of the Stolper-Samuelson theorem.

Important Implications

Factor Increase	Effect on Labor-Intensive Good	Effect on Capital-Intensive Good
Labor Endowment ↑	**Strong Increase**	Decrease
Capital Endowment ↑	Decrease	**Strong Increase**

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Main Mechanism

Suppose **Labor endowment increases** while `goods prices remain constant`:

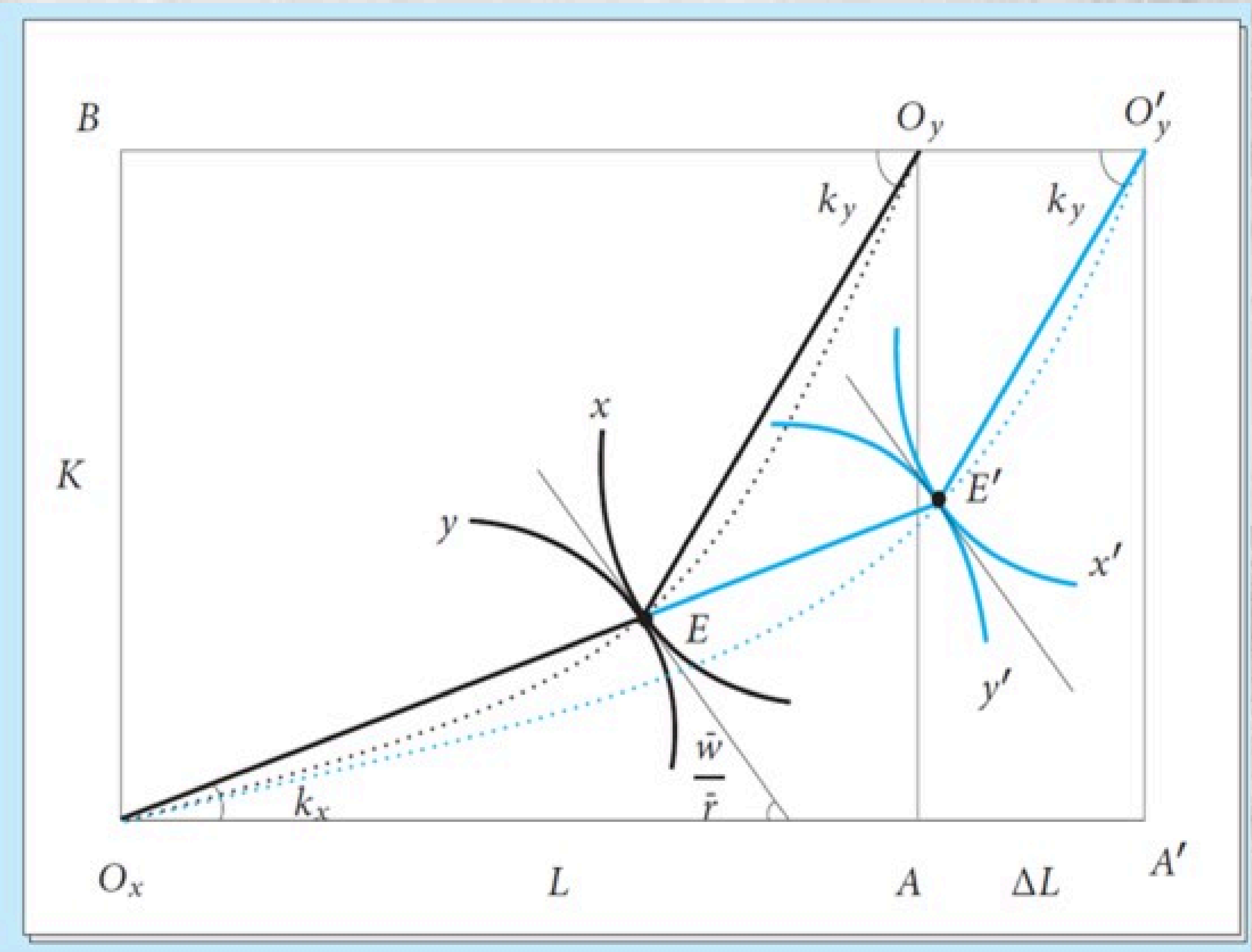
1. At the initial wage-rental ratio (w/r), firms want to use `more labor`.
2. The `labor-intensive` sector (say Good X) can absorb more labor profitably.
3. Resources (especially capital) are pulled away from the capital-intensive sector (Good Y).
4. Result:
   - Output of **labor-intensive good (X)** ↑ significantly
   - Output of **capital-intensive good (Y)** ↓

This is called the **Rybczynski Effect**.

Graphical Explanation

Using the Edgeworth-Bowley Box:

- Increase in labor shifts the box outward horizontally (more labor).
- The efficiency locus (contract curve) shifts.
- At constant factor prices (constant w/r), the new equilibrium point moves such that:
  - Labor-intensive industry expands a lot.
  - Capital-intensive industry contracts.



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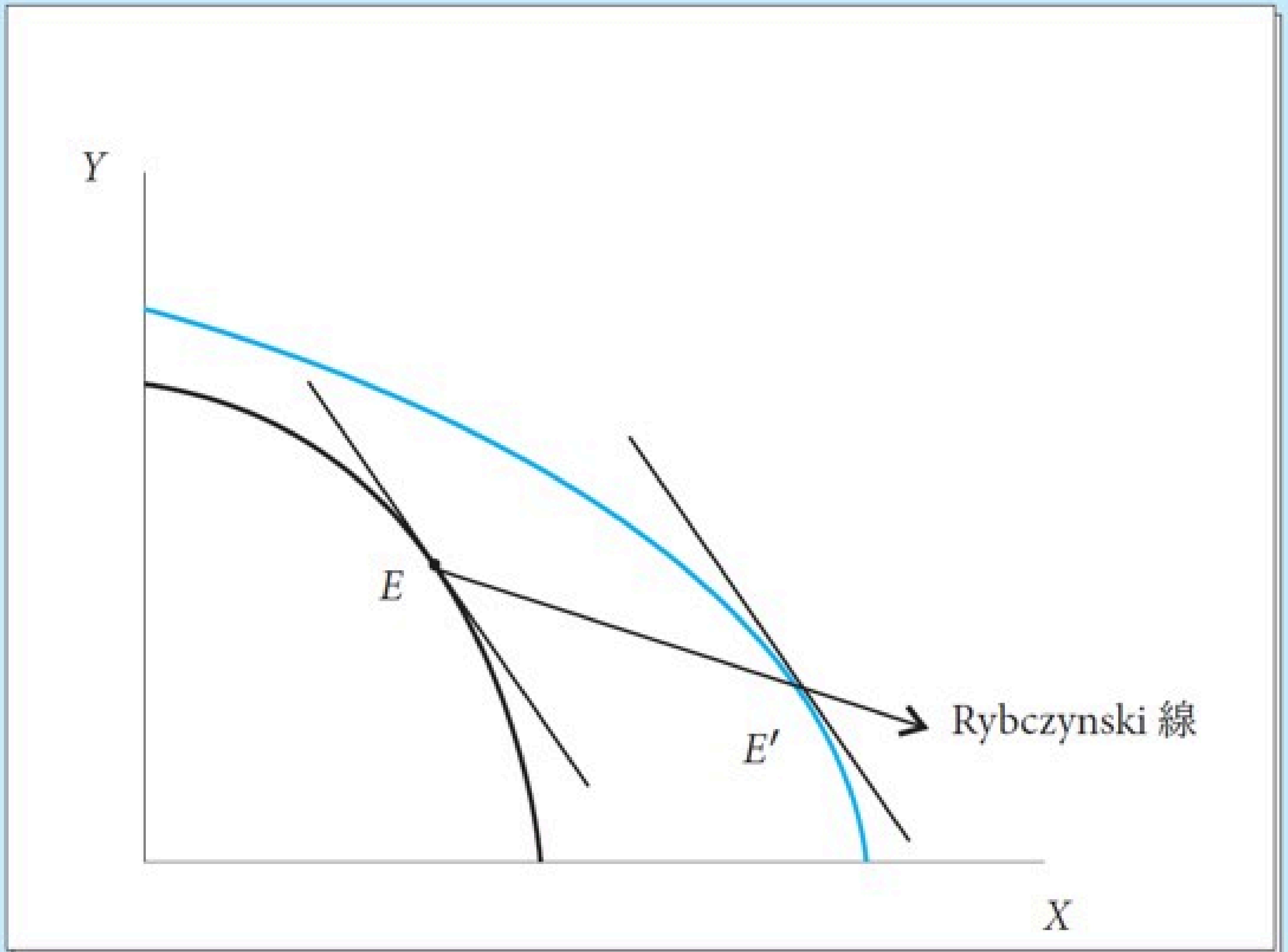
Using Production Possibility Frontier (PPF):

- Increase in labor endowment **shifts the PPF outward**, but **biased** toward the labor-intensive good.
- **At constant relative prices**, production moves **along the new PPF** in a way that increases X a lot and decreases Y.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Political Economy Implications

Context	Rybczynski Prediction
Immigration/Refugee inflows	Labor-intensive sectors (construction, agriculture, services) expand; capital-intensive sectors contract
Foreign Direct Investment (FDI)	Capital inflow expands manufacturing/infrastructure; labor-intensive traditional sectors shrink
Natural resource discovery	Resource-intensive sector booms; other sectors contract (pre-price feedback; distinct from Dutch Disease)
Demographic transition	Aging population (↓L) contracts labor-intensive sectors, relatively expands capital-intensive ones
Industrial policy	Subsidized capital accumulation shifts output toward capital-intensive industries, potentially crowding out others

Empirical Evidence & Limitations

Supportive Evidence:

- **Mariel Boatlift (1980)**: Miami's labor supply shock expanded construction and low-skilled services, consistent with Rybczynski.
- **EU Eastern Enlargement**: Capital flows from West to East expanded capital-intensive manufacturing in accession countries.
- **Chinese regional studies**: Provinces with higher capital accumulation saw disproportionate growth in machinery/electronics output.

Intuitive Explanation

When labor increases **at fixed prices**:

- The economy wants to use the extra labor.
- Since good X uses labor more intensively, expanding X absorbs the new labor efficiently.
- To release more capital (which is needed along with the extra capital in X), the economy must contract industry Y (capital-intensive), freeing up capital.

Link to Trade Patterns (H-O Model)

- A capital-abundant country (higher K/L) will produce relatively more of the capital-intensive good — this is the Rybczynski effect working across countries.
- Explains why endowment differences lead to different production patterns and thus trade patterns (H-O theorem).

Key Takeaways

1. **Endowment changes reshape production structure** even without price changes.
2. **Magnification effect**: Output of the intensive sector grows faster than the factor itself.
3. **Contraction of the non-intensive sector** is a necessary condition for factor price stability.
4. The theorem is a **partial equilibrium/ceteris paribus result** within a general equilibrium framework; in reality, price feedback (via Stolper-Samuelson) and demand adjustments modify outcomes.
5. Highly relevant for immigration policy, FDI strategy, demographic planning, and industrial development.

Mathematical Derivation

1. Full Employment Conditions (Endowment Constraints)

$$a_{LX}Q_X + a_{LY}Q_Y = L$$
$$a_{KX}Q_X + a_{KY}Q_Y = K$$

In matrix form:

$$\mathbf{AQ} = \mathbf{V}$$

where $\mathbf{A}$ is the input coefficient matrix, $\mathbf{Q} = \begin{bmatrix} Q_X \\ Q_Y \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} L \\ K \end{bmatrix}.$

2. Differentiated Form (Percentage Changes)

Differentiate the full employment conditions while holding **goods prices constant** (which implies input coefficients a_{ij} are `fixed` because factor prices are fixed by goods prices via zero-profit conditions):

$$\lambda_{LX}\hat{Q}_X + \lambda_{LY}\hat{Q}_Y = \hat{L}$$
$$\lambda_{KX}\hat{Q}_X + \lambda_{KY}\hat{Q}_Y = \hat{K}$$

Where **allocation shares** (λ_{ij}) are:

- $\lambda_{LX} = \frac{a_{LX} \cdot Q_X}{L}$ = share of labor used in industry X
- $\lambda_{KX} = \frac{a_{KX} \cdot Q_X}{K}$ = share of capital used in industry X (and $\lambda_{LX} + \lambda_{LY} = 1$, etc.)

Because X is capital-intensive, we have $\lambda_{KX} > \lambda_{LX}$ and $\lambda_{KY} < \lambda_{LY}$.

3. Matrix Form and Solution

$$\begin{bmatrix} \lambda_{LX} & \lambda_{LY} \\ \lambda_{KX} & \lambda_{KY} \end{bmatrix} \begin{bmatrix} \hat{Q}_X \\ \hat{Q}_Y \end{bmatrix} = \begin{bmatrix} \hat{L} \\ \hat{K} \end{bmatrix}$$

Let $\mathbf{\Lambda}$ be the matrix above. Its determinant $|\mathbf{\Lambda}| > 0$ because of the intensity ranking.

Solving using Cramer's rule:

$$\hat{Q}_X = \frac{\lambda_{KY}\hat{K} - \lambda_{LY}\hat{L}}{|\mathbf{\Lambda}|} \hat{Q}_Y = \frac{\lambda_{LX}\hat{L} - \lambda_{KX}\hat{K}}{|\mathbf{\Lambda}|}$$

Main Results (Rybczynski Effects)

Case 1: Increase in Capital Endowment ($\hat{K} > 0, \hat{L} = 0$)

- $\hat{Q}_X > \hat{K} > 0$ (output of capital-intensive good rises **more than proportionally**)
- $\hat{Q}_Y < 0$ (output of labor-intensive good **falls**)

Case 2: Increase in Labor Endowment ($\hat{L} > 0, \hat{K} = 0$)

- $\hat{Q}_Y > \hat{L} > 0$ and $\hat{Q}_X < 0$

This is called the **magnification effect** on the quantity side.